Assignment - 2

CSL 2050 - Pattern Recognition & Machine Learning

Max. Marks: 50

Submission Instructions:

- Students must submit one Jupyter notebook (.ipynb) file containing:
  - Executable code
  - Visible outputs (tables, figures, metrics)
  - Brief comments or markdown explanations where required
- File naming format : **RollNo.-Name.ipynb** example, 21CS045-AmitSharma.ipynb
- Do Not submit PDFs, word files, or screenshots.
- Late submissions will be penalized as per course policy.
- Date of last submission: **5th March 2026, 11:59PM**.

Part A: Dimensionality Reduction via PCA

Dataset: Heart Disease Dataset
https://www.kaggle.com/datasets/cherngs/heart-disease-cleveland-uci?select=heart_cleveland_upload.csv

Tasks:

1. Perform exploratory analysis i.e., evaluating data distribution, etc. (use libraries) [5]

2. Implement Principal Component Analysis from scratch **without using any libraries.** Visualize the covariance matrix, eigenvalues (sorted) and explained variance ratio. [5]

3. Reduce the dataset to 2 principle components and visualize the 2D scatter plot. (use libraries) [5]

4. Reconstruct and Error Analysis: reconstruct data from first 2, 3 and 5 components and compute reconstruction error (Mean Squared Error). [5]

5. What happens if the features are not standardized before PCA? [5]

Part B: Linear Discriminant Analysis (LDA) for Supervised Projection

Dataset: Iris Dataset
https://archive.ics.uci.edu/dataset/53/iris

Tasks:

1. Perform exploratory analysis i.e., names of all features, dataset shape, class distribution. (use libraries) [5]

2. Implement Linear Discriminant Analysis (**from scratch by creating a function**) (The implementation must contain computation of overall mean, class-wise mean, within-class and between-class scatter matrices) Visualize the heatmaps of these matrices [10]

3. Project dataset onto first 1 and 2 discriminant components. Plot the scatter diagrams for visualizations. [5]

4. Use projected features to train a simple classifier (e.g., Logistic Regression) and report the accuracy. Compare the accuracies before and after LDA[5]